**This assignment will not be graded and is only for practice.**

## Multiple Select Questions (MSQ):

1) Match the systems of linear equations in Column A with their number of solutions in Column B and their geometric representation in Column C. ***1 point***

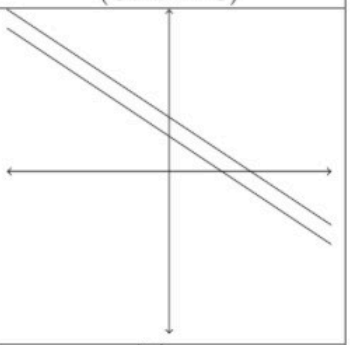
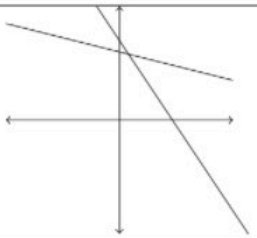
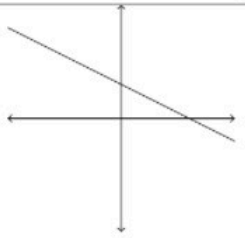
| | System of linear equations (Column A) | | Number of solutions (Column B) | | Geometric representations (Column C) |
|---|---|---|---|---|---|
| i) | $3x + 2y = 7,\ x + 4y = 12$ | a) | Infinite solutions | 1) |  |
| ii) | $2x + 3y = 5,\ 8x + 12y = 13$ | b) | No solution | 2) |  |
| iii) | $x + 2y = 3,\ 4x + 8y = 12$ | c) | Unique solution | 3) |  |

Table: M2W1PT1

- [ ] i) → c) → 2)
- [ ] i) → a) → 1)
- [ ] ii) → c) → 3)
- [ ] iii) → a) → 2)
- [ ] ii) → b) → 1)

No, the answer is incorrect.

Score: 0

Accepted Answers:

i) → c) → 2)
ii) → b) → 1)

2) Choose the set of correct options **1 point**

- [ ] If both $A$ and $B$ are $2 \times 2$ real matrices and $det(AB) = 0$, then $det(A) = 0$ or $det(B) = 0$.
- [ ] If $A$ is a $3 \times 3$ real matrix with non-zero determinant and $k$ is some real number, then $det(k\,A) = k^3 \times det(A)$.
- [ ] If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$, then $A^{10} = 2^{10} \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$ (where $A^n$ is the matrix $A \times A \times \ldots \times A$, $n$-times).
- [ ] The number of scalar additions to be done to compute the matrix $AB$, where $A$ is a $3 \times 2$ matrix and $B$ is a $2 \times 3$ matrix, is 9.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If both $A$ and $B$ are $2 \times 2$ real matrices and $det(AB) = 0$, then $det(A) = 0$ or $det(B) = 0$.
If $A$ is a $3 \times 3$ real matrix with non-zero determinant and $k$ is some real number, then $det(k\,A) = k^3 \times det(A)$.
If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$, then $A^{10} = 2^{10} \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$ (where $A^n$ is the matrix $A \times A \times \ldots \times A$, $n$-times).
The number of scalar additions to be done to compute the matrix $AB$, where $A$ is a $3 \times 2$ matrix and $B$ is a $2 \times 3$ matrix, is 9.

3) Choose the set of correct options **1 point**

- [ ] A triangular $3 \times 3$ matrix has non-zero determinant if and only if all the diagonal entries are non-zero.
- [ ] If $A$ and $B$ are $3 \times 3$ matrices then $det(A + B) = det(A) + det(B)$.
- [ ] If $A$ is a $3 \times 3$ matrix and $B$ is a matrix obtained from $A$ by multiplying each column of $A$ by its column number, then $det(B) = 6det(A)$.

☐ If the sum of the first and the third row vectors of a $3 \times 3$ matrix A is equal to the second row vector of $A$, then $det(A) = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A triangular $3 \times 3$ matrix has non-zero determinant if and only if all the diagonal entries are non-zero.
If $A$ is a $3 \times 3$ matrix and $B$ is a matrix obtained from $A$ by multiplying each column of $A$ by its column number, then $det(B) = 6det(A)$.
If the sum of the first and the third row vectors of a $3 \times 3$ matrix A is equal to the second row vector of $A$, then $det(A) = 0$.

4) Mahesh bought 2 kg potato and $c$ kg dal from a shop, and paid ₹ 200 to the shopkeeper. Gaurav bought 4 kg potato and 4 kg dal, and paid ₹ $d$ to the shopkeeper. If $x_1$ represents the price of 1 kg potato and $x_2$ represents the price of 1 kg dal, then choose the set of correct options. ***1 point***

☐ The matrix representation to find $x_1$ and $x_2$ is $\begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$

☐ The matrix representation to find $x_1$ and $x_2$ is $\begin{bmatrix} 2 & c \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$

☐ If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$, then the price of 1 kg potato that he thus arrives at, will not be unique.

☐ If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$, then the price of 1 kg potato that he thus arrives at, will be unique.

☐ If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d = 400$, then he will be able to find the price (as a numerical value) of 1 kg potato.

☐ If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d = 400$, then he will fail to find the price (as a numerical value) of 1 kg potato.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The matrix representation to find $x_1$ and $x_2$ is $\begin{bmatrix} 2 & c \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$

If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$, then the price of 1 kg potato that he thus arrives at, will not be unique.

If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d = 400$, then he will fail to find the price (as a numerical value) of 1 kg potato.

5) The marks obtained by Safina, Ram and Pratiksha in Quiz 1, Quiz 2 and End sem (with the maximum marks for each exam being 100) are shown in Table W1PT2. ***1 point***

| | Quiz 1 | Quiz 2 | End sem |
|---|---|---|---|
| Safina | 89 | 95 | 88 |
| Ram | 92 | 81 | 98 |
| Pratiksha | 85 | 93 | 98 |

Table: W1PT2

The weightage of marks in final grade(in percent) of Quiz 1, Quiz 2, and End sem is shown in Table W1PT3.

| | In percent (%) |
|---|---|
| Quiz 1 | 20 |
| Quiz 2 | 20 |
| End sem | 60 |

Table: W1PT3

Choose the set of correct options.

Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix:

☐ $\begin{bmatrix} 93.4 \\ 94.4 \\ 89.6 \end{bmatrix}$

Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix :

☐ $\begin{bmatrix} 89.6 \\ 93.4 \\ 94.4 \end{bmatrix}$

☐ The order of the matrix which represents final grades(in 100) of Safina, Ram and Pratiksha is $3 \times 1$

If bonus marks given to Safina, Ram and Pratiksha are represented by the following matrix $\begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix}$ then the overall final grades (in 100) can be

☐ represented by the matrix:

$\begin{bmatrix} 91 \\ 96 \\ 94.4 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix :

$\begin{bmatrix} 89.6 \\ 93.4 \\ 94.4 \end{bmatrix}$

The order of the matrix which represents final grades(in 100) of Safina, Ram and Pratiksha is $3 \times 1$

If bonus marks given to Safina, Ram and Pratiksha are represented by the following matrix $\begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix}$ then the overall final grades (in 100) can be represented by the matrix:

$$\begin{bmatrix} 91 \\ 96 \\ 94.4 \end{bmatrix}$$

**Numerical Answer Type (NAT):**

6) Let $A$ be a $3 \times 3$ matrix with non-zero determinant and $B$ be a matrix obtained by adding 5 times of first row of$A$ to the third row of $A$ and adding 10 times of second row of $A$ to the first row of $A$.What is the value of $det(2AB^{-1})$?

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

***1 point***

7) If $A = \begin{bmatrix} 4 & 6 & 10 \\ 1 & 4 & -5 \\ 4 & 1 & 10 \end{bmatrix}$ ,then find $det(A)$.

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -150

***1 point***

**Comprehension Type Question:**

**(Use the below information for question 8, 9 and 10)**

A shopkeeper sells three types of clothes- shirts, jeans, and T- shirts- in three different sizes: small, medium, and large. In a week, he sold 1 small, 1 medium and 2 large sized shirts; 2 small, $c$ medium and 6 large sized jeans, and 1 small, 3 medium and $c - 5$ large sized T-shirts (where $c$ is an integer). The price of shirts, jeans, and T-shirts remain the same for different sizes (i.e., small, medium, and large sized shirts have the same price; similarly, small, medium, large sized jeans have the same price; and small, medium, large sized T-shirts have same price). The shopkeeper earned ₹7 , ₹27 and ₹43 (in thousand) in that week, for small, medium, and large sized clothes respectively.

8) Let $s, j, t$ represents the price (in thousand) of 1 shirt, 1 jeans and 1 T-shirt respectively and we want to find $s, j, t$ by solving a system of linear equations represented by the matrix form $Ax = b$, where $x = (s, j, t)^T$ and $b = (7, 27, 43)^T$. If sum of entries in the first row of $A$ is $p$, sum of entries in the second row of $A$ is $q$ and sum of entries in the third row of $A$ is $r$ (take the topmost row as the fist row ), then find the value of $p + q - r$.

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

***1 point***

9) If $det(A) = 122$, then how many medium sized jeans were sold in the week?

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 16

***1 point***

10) If $A$ is the matrix as above (in question 8) and $det(A) = 122$, then what is the price(in thousand) of a T-shirt, where price (in thousand) of a shirt is 2 and price (in thousand) of a jeans is 1?

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

***1 point***

**Your score is: 0/10**